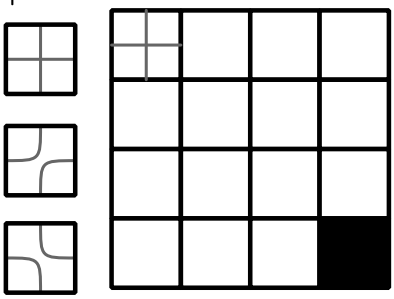
MathsJam Shout

October 2026

Perth MathsJam, UK

Play
Black
a pencil and paper game

Draw a square or rectangular grid (8-by-8 works well, but you can choose any size you want!) Shade the bottom-right corner (this is the square you are trying to get to) and draw a cross in the top left.

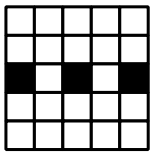


On your turn you may only draw one of the following three shapes in one of the squares on the grid. At least one part of the mark has to join onto the path that is being made. Note that in the cross shape, the path goes straight across, so the path is smooth (no right-angle turns).

If a player joins the path on to the edge of the board they lose . . . but if they join the path to the shaded square they win.

Puzzle
Clear the Board

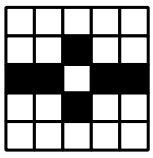
Draw a 5 by 5 grid. You will need counters, but if you don't have any then just cut out lots of shaded squares to use as counters. Place three counters as follows:



Now choose a square to change (from shaded to unshaded, or vice versa) and also change the squares above, below, to the left and to the right (i.e. a cross shape).

Now keep choosing squares and changing the cross shape until you clear the board.

e.g. changing middle square:



We have provided 12 starting patterns in order of difficulty, but you can easily make your own by starting with a blank board and applying the cross-shaped changes a few times to leave a pattern to give to a fellow MathsJammer to solve.

Sheet of starting positions: mathsj.am/clear-the-board

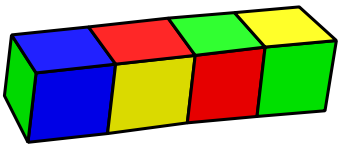
Problem
Half
a non-calculator puzzle

Using the digits 1 to 9 each once, and only once, make as many fractions equal to a half as possible:

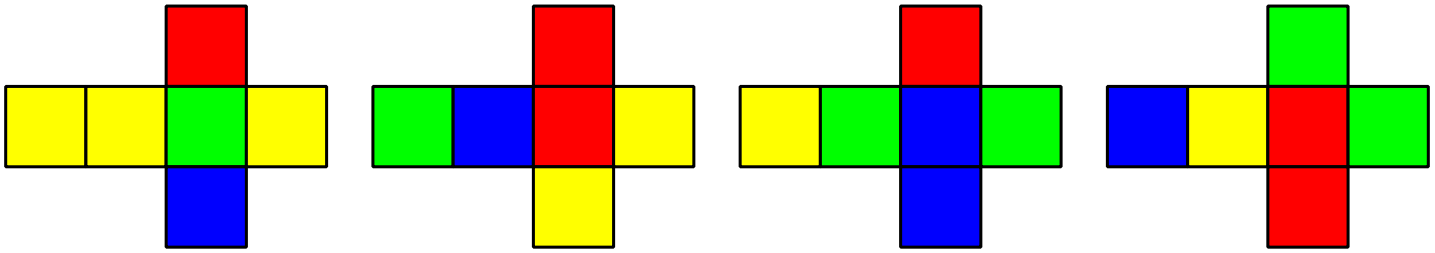
$$\frac{9273}{18546} = \frac{1}{2} \qquad \frac{9327}{18654} = \frac{1}{2}$$

Can you be sure you have found all of them? No calculating devices or computer coding allowed!

Make
Instant Insanity Puzzle



From the following nets, cut out and make four pretty cubes. Now place the four cubes so they make a 4-by-1-by-1 cuboid, with each of the 4-by-1 sides showing each colour once only (the 1-by-1 ends don't matter).



MathsJam Shout is a monthly sheet of ideas for activities to do at a MathsJam night. It's created using suggestions from a different MathsJam each month, and if you'd like to submit suggestions for a month in the future, email **katie@mathsjam.com**

Maths Jam is a monthly opportunity for like-minded self-confessed maths enthusiasts to get together in a pub and share stuff they like. Puzzles, games, problems, or just anything they think is cool or interesting. Monthly Maths Jam nights happen in over 70 locations around the world, on the second-to-last Tuesday of each month. To find your nearest Maths Jam, visit the website at **mathsjam.com.**